# Do Now · retrieval

## Paging, segmentation and virtual memory

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is RAM used for?

### Q2

What are remote virtual storage files?

### Q3

Why might a process have to wait on memory I/O?

### Q4

A 4-row, 3-column zero-indexed array: give its final valid index pair and number of values.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Understand — Pages and frames have equal fixed sizes; the page table maps virtual pages to physical frames.

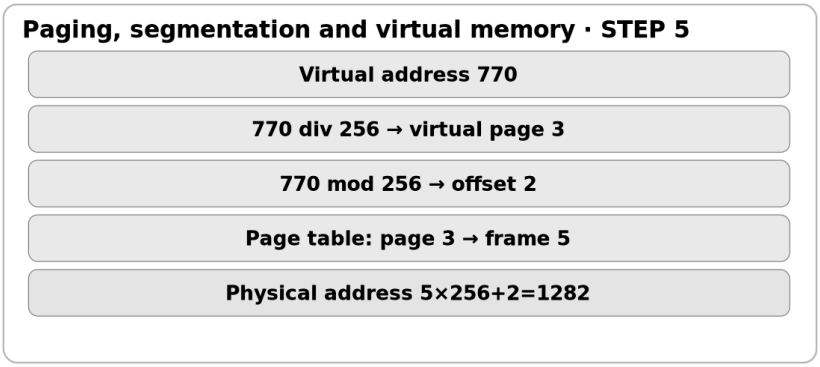
## Core idea 2

Trace — Divide the virtual address by page size for the page number; use remainder as the offset.

## Core idea 3

Explain — A page fault requires a non-resident page to be loaded. Variable-sized segmentation has different fragmentation characteristics.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

page — Virtual fixed-size memory block

frame — Physical fixed-size RAM block

page fault — Page absent from RAM

segmentation — Logical variable-length regions

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

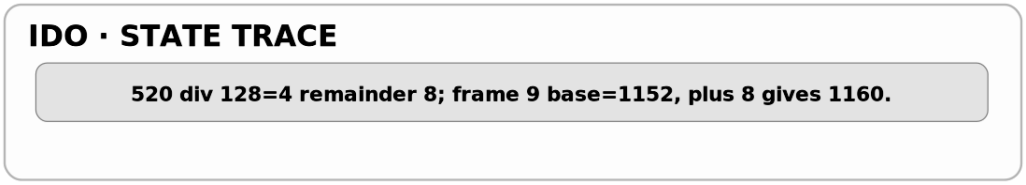
## Problem

Translate virtual address 520 with page size 128 and page 4 mapped to frame 9.

## Follow the worked method

1. Predict first — 520 div 128=4 remainder 8; frame 9 base=1152, plus 8 gives 1160.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first
615 div 200=3 remainder 15. Page 3 maps to frame 7, hence physical address 1415.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Define page, frame, page table and page fault. [4]

## Question 2

Page size 256, VA 770, page 3 mapped to frame 5. Find physical address. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Compare paging and segmentation and explain thrashing. [6]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

Paging changes the offset within the page when it changes frame.

## Use key vocabulary from memory

1. Explain page in your own words.

2. Explain frame in your own words.

3. Explain page fault in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.